



A. Adhoc King

CPU: 1.00s

Memory: 2048MB

Pebae has an integer n . For an integer sequence $a_1, a_2, \dots, a_n$ and an integer k , she defines $f(k)$ as $(a_1 \times a_2 \times \dots \times a_k) \& (a_1 + a_2 + \dots + a_k)$ — that is, $f(k)$ is the **bitwise AND** of the product and sum of the first k elements of the sequence.

An integer sequence is good if $f(i) \neq f(j)$ is satisfied over all pairs $1 \leq i < j \leq n$.

Now, Pebae wants to find a good sequence of length n such that all elements of the sequence are from 2 to $3n$ and the difference between the maximum and minimum element of the sequence is at most 10.

Help Pebae find such a sequence. It can be shown that such a sequence always exists under the given constraints.

Input Description

The first line contains a single integer $t (1 \leq t \leq 10^4)$ — the number of test cases.

The first and only line of each test case contains an integer $n (2 \leq n \leq 10^6)$.

It is guaranteed that the sum of n over all test cases does not exceed 10^6 .

Output Description

For each test case, print n integers — the integer sequence that satisfies the conditions mentioned in the statement. If there are multiple such sequences, output any.

Samples

Input

Output

3
3
2
65 7 5
3 2
11 12 3 2 9 12

Note

In the first test case,

- $f(1) = 5 \ \& \ 5 = 5$.
- $f(2) = (5 \times 7) \ \& \ (5 + 7) = 35 \ \& \ 12 = 0$.
- $f(3) = (5 \times 7 \times 5) \ \& \ (5 + 7 + 5) = 175 \ \& \ 17 = 1$.

Here, all the values of $f(k)$ are distinct, so the sequence is good.

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Ends in
03:34:14

THE CONTEST IS RUNNING

SUBMIT

Language

C++

-O2 -static -std=gnu++17

SELECT SOURCE

SUBMIT

MY SUBMISSIONS



No submissions yet!

